End-to-End Retail Performance and Behavioral Analytics

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Introduction:

Company Name: RetailNova Inc.

Industry: Retail & E-Commerce

Founded: 2016

Headquarters: Chicago, USA

Sales Channels: Online platform, 60+ physical stores across 8 regions.

RetailNova Inc. is a mid-sized retail company that sells a wide range of consumer products including electronics, apparel, home & kitchen, and personal care items. The company operates both online and offline channels, giving customers the flexibility to shop from anywhere.

Over the last few years, RetailNova has experienced rapid growth in online orders, especially during the pandemic.

2. Problem Statement:

However, recent quarters have seen increased product returns, customer churn, and declining profitability in certain regions and product categories.

To maintain its market position and plan for the next growth phase, RetailNova's leadership wants to become more data-driven in its decision-making.

3. Objectives:

The primary objectives of this project are:

- Identify key sales trends and anomalies
- Understand customer behavior and segmentation
- Optimize product performance and inventory
- Improve store and regional profitability
- Support decision-making through data visualization and reporting

4. Methodology:

The methodology outlines the step-by-step process followed in the project, from data collection to final analysis and visualization. It ensures a structured approach to achieve the project's objectives by using various tools and techniques.

Data Cleaning:

The First step is **data cleaning**. The goal of this phase is to ensure the dataset is free from errors, missing values, and irrelevant information.

Cleaning Steps:

- Removing the Duplicates
- Replacing the missing values with relevant placeholders (Eg:Mean)
- Adding the new columns like Profit and Age Group
- Run the basic EDA(Exploratory Data Analysis) using python

Tools Used: Jupyter Notebook

SQL Analysis:

With a clean dataset, the next step is analysis using Sql, Through SQL, organizations can track performance, discover trends, understand customer behavior, and identify what drives profits or causes losses.

Tools Used: MS SQL

Key Metrics are :

- Total Profit

	Total_Profit
1	243772.19

- Return %

	Total Return %
1	6.65

- Total Discount %

	Total Discount %
1	8

Sql used to solve the business questions:

1. What is the total revenue generated in the last 12 months?

	total_revenue_last_12_months
1	121805.869433403

2. Which are the top 5 best-selling products by quantity?

Results | Messages

	product_id	product_name	total_quantity_sold
1	P0820	BrandC House	40
2	P0053	BrandC Decision	33
3	P0336	BrandA Feel	33
4	P0874	BrandB Stop	32
5	P1123	BrandB Choice	27

3. How many customers are from each region?

	region	total_customers
1	West	204
2	East	201
3	North	199
4	South	196

4. Which store has the highest profit in the past year?

	Store_Name	Profit
1	Ortiz-Smith Store	4553.53

5. What is the return rate by product category?

	Category	Return_Rate
1	Personal Care	6.85 %
2	Apparel	6.45 %
3	Electronics	6.19 %
4	Home & Kitchen	7.11 %

6. What is the average revenue per customer by age group?

	Age_Group	Avg_Revenue
1	18-25	281.65
2	15-18	253.26
3	26-35	309.25
4	50+	248.57
5	36-50	275.12

7. Which sales channel (Online vs In-Store) is more profitable on average?

	sales_channel	avg_profit_per_order
1	In-Store	83.1463815797945
2	Online	76.9052424153748

8. How has monthly profit changed over the last 2 years by region?

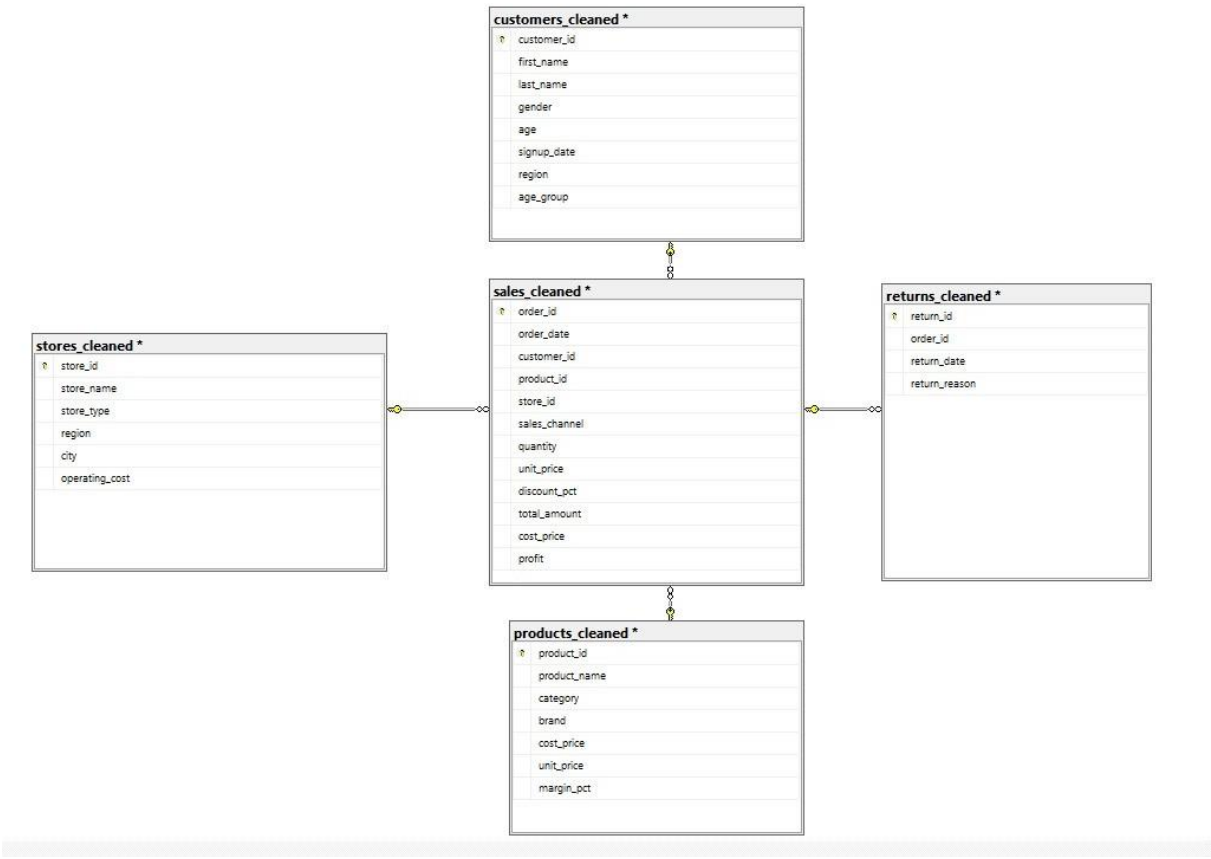
	Region	Sales_Year	Sales_Month	Monthly_Profit
1	East	2023	4	864.21
2	East	2023	5	1139.04
3	East	2023	6	1442.83
4	East	2023	7	1512.87
5	East	2023	8	3440.97
6	East	2023	9	1354.46
7	East	2023	10	1693.88
8	East	2023	11	797.29
9	East	2023	12	1395.33
10	East	2024	1	818.32
11	East	2024	2	1168.21
12	East	2024	3	1001.62

9. Identify the top 3 products with the highest return rate in each category.

	Category	Product_Name	Return_Rate
1	Apparel	BrandA Other	100%
2	Apparel	BrandB Finally	100%
3	Apparel	BrandB Image	100%
4	Electronics	BrandC Let	100%
5	Electronics	BrandC Ever	100%
6	Electronics	BrandD Executive	100%
7	Home & Kitchen	BrandD Official	100%
8	Home & Kitchen	BrandC Today	100%
9	Home & Kitchen	BrandC Rate	100%
10	Personal Care	BrandD What	100%
11	Personal Care	BrandD Knowledge	100%
12	Personal Care	BrandB Line	100%

10. Which 5 customers have contributed the most to total profit, and what is their tenure with the company?

	Customer_Name	Profit	Tenure (in months)
1	Kathy Winters	3861.36	59 Months
2	Terry Davis	3385.17	34 Months
3	Christopher Kennedy	3206.17	61 Months
4	Levi Bradford	2697.7	24 Months
5	James Fischer	2523.61	37 Months



Data Visualization:

It helps people quickly understand patterns, trends, and relationships that might stay hidden in raw data.

Data visualization also makes communication easier, turning complex insights into simple stories that everyone in a business can understand and act on. In short, data visualization transforms data into a visual language that drives smarter, faster, and more confident decisions.

Tools Used: Power BI

Creating an interactive dashboard that includes:

1. Sales Overview Report
2. Customer Insights Report
3. Product Performance Report
4. Store Analysis Report
5. Return Analysis Report

Reports:

Product Performance Report

Total Profit

$320.75K

Total Quantity Sold

5454

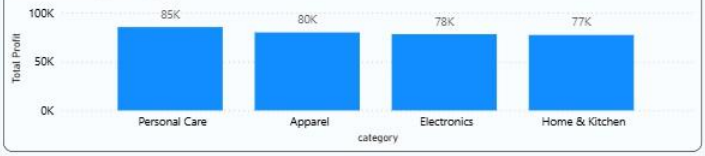
Return Rate

26.60%

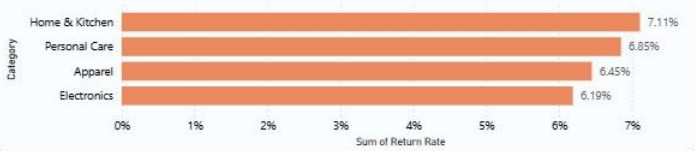
Sales Quantity vs Product vs Region

product_name	East	North	South	West	Total
BrandA A		4			4
BrandA Ability		1	2		3
BrandA About	1	2	1		4
BrandA Across	2	1	2	3	8
BrandA Activity	1		4	2	1	8
BrandA Actually	3				3
BrandA After	1			1	2
BrandA Agree	3	2		3	1	9
BrandA Although	1		5	1		7
BrandA Among				14	14
BrandA And	1				1
BrandA Argue	5				5
BrandA Arm	2				2
BrandA Arrive		1	1	3	5
BrandA Author			1	1	2
BrandA Baby	2			1	3
Total	1756	666	884	870	1278	5454

Total Profit by Category



Return Rate by Category



Store Analysis Report

Total Sales

818.18K

Total Profit

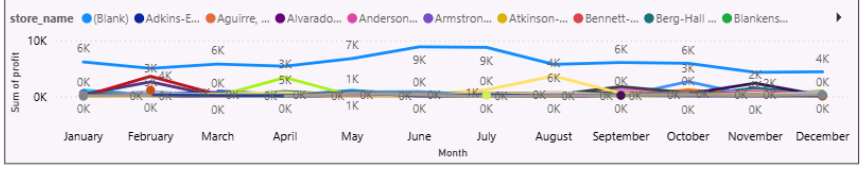
$320.75K

Store with Highest Sales

145

Robinson PLC Store

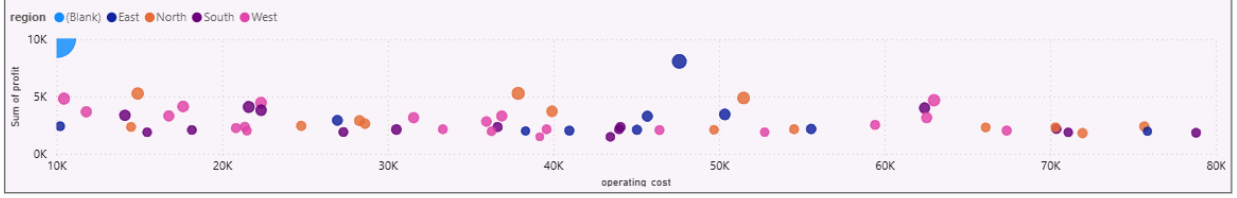
Monthly Profit by Store

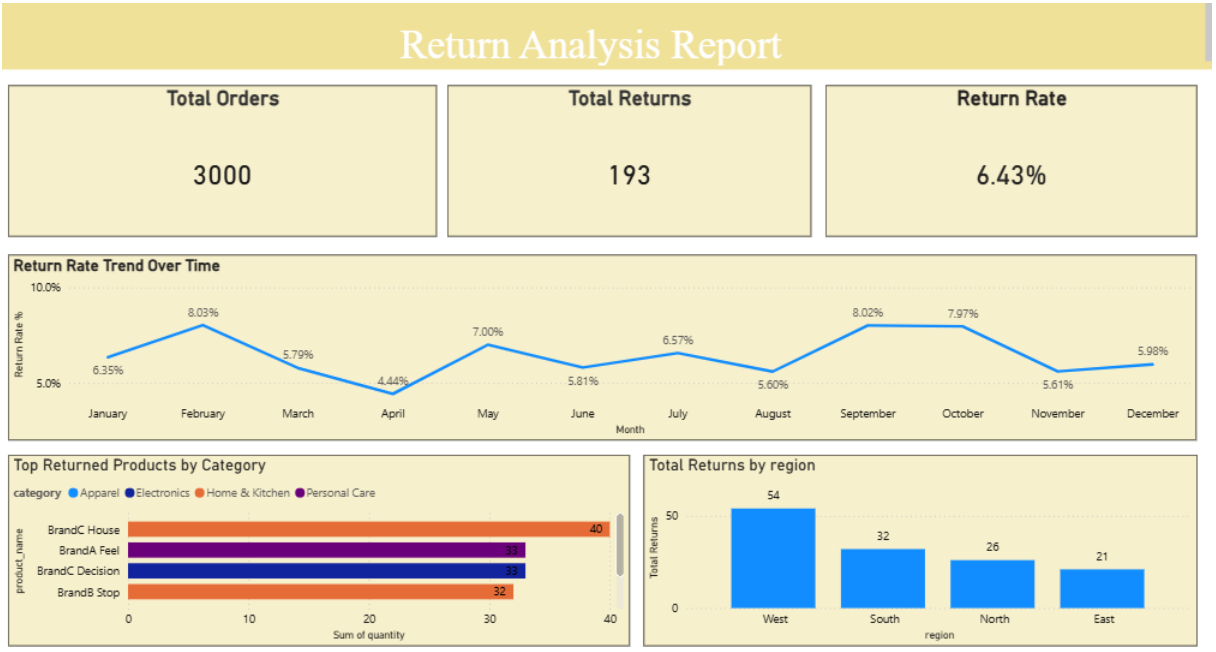


Store with Highest profit in past year.

Ortiz-Smith Store

Store Profitability (Profit vs Operating Cost)





5.Key Findings:

1. Store sales are higher than online sales, making offline channels the primary revenue source.
2. Middle-aged customers (36–55 years) contribute the maximum share of total revenue.
3. Home & Kitchen category has the highest return rate, increasing product returns.
4. Some stores generate high sales but low profit due to higher operating costs.
5. Returns are concentrated in the west region, indicating regional performance issues.

6. Key Insights:

1. Expanding and improving online sales channels can significantly increase total revenue.
2. Targeted marketing and loyalty programs for middle-aged and high-value customers can improve customer retention.
3. Reducing product return reasons (quality, size mismatch, unclear descriptions) will directly improve profitability.
4. Optimizing operating costs is essential to convert high sales into higher profit at the store level.
5. Region-specific and category-specific strategies can help reduce returns and improve overall business performance.

7. Challenges Faced:

Store–Sales Relationship Error

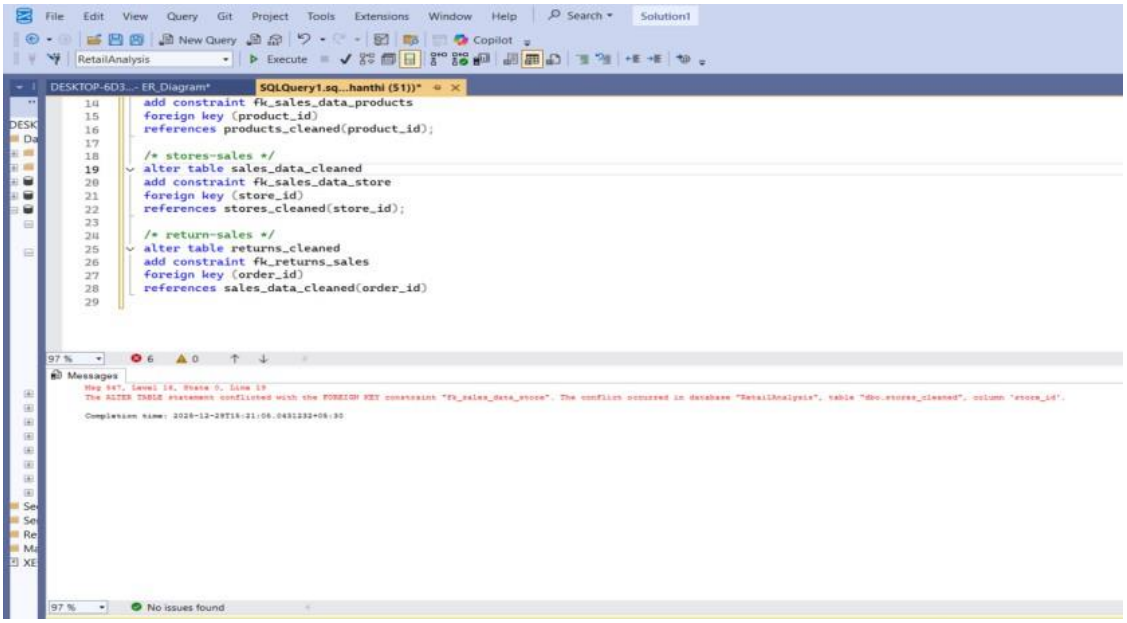
I tried to create a foreign key relationship between:

sales_cleaned(store_id)→ stores_cleaned(store_id)

using:

```
ALTER TABLE sales_cleaned
ADD CONSTRAINT fk_sales_storess
FOREIGN KEY (store_id)
REFERENCES stores_cleaned(store_id);
```

SQL Server did not allow the foreign key creation.



Reason:

Sales table contained a value that did not exist in the Stores table.

- sales_cleaned.store_id = 'online_store'
- But online_store was missing in stores_cleaned
- Foreign key rule:
- Every value in the child table must already exist in the parent table

Solution:

Fixed the issue by inserting a dummy / logical store record:

INSERT INTO stores_cleaned VALUES ('-', 'online_store', 'online', '-', '-', 0);

After Fix: Foreign Key Created Successfully

8. Conclusion:

The analysis reveals that data-driven decision-making is essential for optimizing sales Trends ,customer behaviour ,product performance and improves store and regional profitability.

9. References:

- **Data Sources:** customers, sales, products, returns, stores csv files.
- **Tools Used:** Python(Jupyter Notebook) , Power BI , Sql.